

$$A = \begin{pmatrix} 1 & Z \\ 0 & 1 \end{pmatrix} \quad \text{---} \overline{\underline{Z}} \text{---}$$

$$A = \begin{pmatrix} 1 & 0 \\ \frac{1}{Z} & 1 \end{pmatrix} \quad \text{---} \overline{\underline{\frac{1}{Z}}} \text{---}$$

$$|U|(\omega) = \frac{U_q}{\sqrt{\alpha^2 + j\beta^2}}$$

Betragsgang

$$\varphi_u(\omega) = \varphi_q - \tan^{-1} \left(\frac{j\beta}{\alpha} \right)$$

Phasengang

$$f(t) = A_0 + \sum_{n=1}^{\infty} (A_n \cos n\omega_1 t + B_n \sin n\omega_1 t) \quad A_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$\omega T = 2\pi$$

$$A_n = \frac{2}{T} \int_0^T f(t) \cos n\omega_1 t dt, \quad B_n = \frac{2}{T} \int_0^T f(t) \sin n\omega_1 t dt.$$

$$(208) \int x \cdot \sin(ax) dx = \frac{\sin(ax)}{a^2} - \frac{x \cdot \cos(ax)}{a}$$

$$(232) \int x \cdot \cos(ax) dx = \frac{\cos(ax)}{a^2} + \frac{x \cdot \sin(ax)}{a}$$

$$(205) \int \sin^2(ax) dx = \frac{x}{2} - \frac{\sin(2ax)}{4a} = \frac{x}{2} - \frac{\sin(ax) \cdot \cos(ax)}{2a}$$

$$(229) \int \cos^2(ax) dx = \frac{x}{2} + \frac{\sin(2ax)}{4a} = \frac{x}{2} + \frac{\sin(ax) \cdot \cos(ax)}{2a}$$

$$\sqrt{j} = \frac{1+j}{\sqrt{2}} \quad \underline{U}_N = \frac{\underline{U}_1 \underline{Y}_1 + \underline{U}_2 \underline{Y}_2 + \underline{U}_3 \underline{Y}_3}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3} \Rightarrow \text{Sternpunktspannung}$$

Leitungstheorie

$$\text{Allg:} \quad Z_{1,KS} = Z_W \cdot \tanh(\gamma \cdot l) \quad Z_{1,LL} = Z_W \cdot \coth(\gamma \cdot l)$$

$$\text{Verlustfrei:} \quad Z_{1,KS} = j \cdot Z_W \cdot \tan(\beta \cdot l) \quad Z_{1,LL} = -j \cdot Z_W \cdot \cot(\beta \cdot l)$$

$$\beta = \omega \sqrt{L'C'}$$

Niedrige Frequenzen (Telefon...):
 $R \gg \omega L', \quad G \ll \omega C'$

$$\alpha = \beta = \sqrt{\frac{\omega \cdot R' \cdot C'}{2}}$$

$$\underline{Z}_W = (1+j) \sqrt{\frac{R'}{2 \cdot \omega \cdot C'}}$$

$$Z_W = \sqrt{\frac{j\omega L' + R'}{j\omega C' + G'}}$$

$$\lambda = \frac{2\pi}{\beta}$$

$$\gamma = \sqrt{j \cdot \omega \cdot R' \cdot C'} = \sqrt{j} \sqrt{\omega \cdot R' \cdot C'} = \frac{(1+j)}{\sqrt{2}} \sqrt{\omega \cdot R' \cdot C'}$$

$$v = \sqrt{\frac{1}{L'C'}}$$

$$Z_W = \sqrt{Z_{KS} \cdot Z_{LL}}$$

$$Z_W = \sqrt{\frac{L'}{C'}}$$

$$v = \frac{\omega}{\beta}$$

$$v = f \cdot \lambda$$

$$\Gamma = \frac{z_2 - z_w}{z_2 + z_w}$$

$$\gamma = \sqrt{(R' + j\omega L') \cdot (G' + j\omega C')} = \alpha + i\beta$$

$$\beta =: \text{Phasenkonstante}$$

$$\gamma =: \text{Ausbreitungskonstante}$$

$$\alpha =: \text{Dämpfungskonstante}$$

$$S = U \cdot I^*$$

$$\varphi = \varphi_u - \varphi_i$$

$$x_{1/2} = -\frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - q}$$

$$\frac{P}{S} = \cos \varphi = \lambda$$

Symmetrisch:

$$P = U \cdot I^* \cdot \cos \varphi$$

$$S_{GES} = S \cdot \sqrt{3}$$